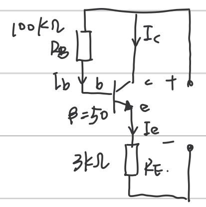
3.2.2

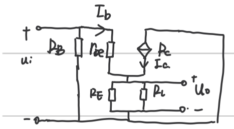
(1) 静态工作点：



$$\begin{cases} U_{cc} = I_b \cdot R_B + U_{BE} + I_e \cdot R_E \\ I_c = \beta I_B \\ I_B + I_c = I_e \\ U_{cc} = U_{cE} + I_e R_E \end{cases}$$

$$\Rightarrow \begin{cases} I_B = 44.66\,\mu A \\ I_c = 2.23\,mA \\ U_{cE} = 5.17V \end{cases}$$

(2)



$$r_{be} = \cancel{200} + (1+\beta)\frac{\cancel{26}}{I_E} = 783\,\Omega.$$

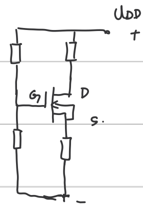
$$A_u = \frac{I_E(R_E // R_L)}{I_b r_{be} + I_E(R_E // R_L)} = 0.987.$$

$$r_i = \frac{u_i}{I_B} = \frac{u_i}{I_b + \frac{u_i}{R_B}} = \frac{I_b r_{be} + I_E(R_E // R_L)}{I_b + \frac{I_b r_{be} + I_E(R_E // R_L)}{R_B}} = 38.3\,k\Omega.$$

$$r_o = R_E // \frac{r_{be}}{1+\beta} = 15.3\,\Omega.$$

3.3.4

(1)



$$U_G = U_{DD}\frac{R_2}{R_1+R_2}. \qquad U_S = R_S I_D.$$

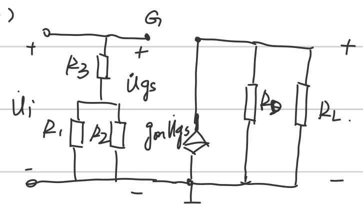
$$I_D = I_{DSS}\left(1 - \frac{U_{GS}}{U_{GS(off)}}\right)^2$$

$$U_{GS} = U_G - U_S = -0.0943\,V$$

$$I_D = 0.4245\,mA.$$

$$U_{DS} = U_{DD} - I_D(R_D + R_S) = 5.57V.$$

(2)



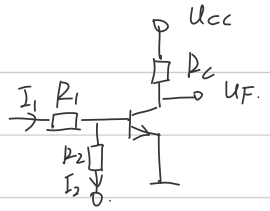
$$A_u = -g_m \cdot \frac{R_D R_L}{R_D + R_L} = -6.35.$$

(3)

$$r_i = R_3 + \frac{R_1 R_2}{R_1 + R_2} = 2.075\,M\Omega.$$

$$r_o = R_D = 22\,k\Omega.$$

3.4.3



$$I_{CS} = \frac{U_{CC} - U_{CES}}{R_C} = 12mA.$$

$$I_{BS} = I_{CS}/\beta = 0.12mA$$

$$I_B \geq I_{BS}$$

$$I_B = I_1 - I_2 = \frac{U_B - U_{BE}}{R_1} - \frac{U_{BE} - (-U_{BB})}{R_2} \geq I_{BS}.$$

$$\therefore U \geq 8.45V.$$

4.1.1

(1) $(A+B)(\bar A + C)(B+C) = (A+B)(\bar A B + \bar A C + CB + CC) = (A+B)(\bar A B + \bar A C + C) = (A+B)(\bar A B + C)$

$$= \cancel{A\bar A B} + AC + \cancel{B\bar A B} + BC = \bar A B + AC + BC = \bar A B + AC + BC + A\bar A$$

$$= (\bar A + C)(A+B)$$

(2)

$$A\bar B + BD + D(A+\bar A) + CDE = A\bar B + D + CD\bar E = A\bar B + D.$$

(3)

$$= ABC(C+\bar C) + \bar A\bar B + \bar A BC = ABC + AB\bar C + \bar A\bar B + \bar A BC = BC(A+\bar A) + \bar A\bar B + AB\bar C$$

$$= AB\bar C + \bar A\bar B + BC.$$

(4)

$$\overline{A\bar B + B\bar C + C\bar A} = ABC + \bar A\bar B\bar C$$

$$\overline{A\bar B + B\bar C + C\bar A} = \overline{A\bar B}\cdot\overline{B\bar C}\cdot\overline{C\bar A} = (\bar A + B)(\bar B + C)(\bar C + A) = (\bar A\bar B + \bar A C + BC)(\bar C + A)$$

$$= \bar A\bar B\bar C + ABC. \quad (\bar A A = 0).$$